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Syndromic surveillance using laboratory test requests: A practical guide informed by experience with two systems

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ABSTRACT

Syndromic surveillance systems can enhance early disease warning, endemic disease monitoring, or help to accumulate proof of disease freedom. In order to provide immediate feedback to achieve these goals, the health data sources scanned should be acquired continuously, in an automated fashion, and should be stored electronically. Recognizing that data from diagnostic test requests often meet these requirements, two systems designed to automatically extract surveillance information from animal laboratory databases have been developed and are described in this paper. These systems are designed to contribute to early disease detection, as well as the timely management of epidemiological information, in a province of Canada and in Sweden, the areas served by the diagnostic laboratories concerned. Classifying in-coming requests into syndromes, the first step, was the most time-consuming and the least portable step between the two systems. The remaining steps were more easily adjusted from one system to implementation in the other. These steps included: retrospective evaluation of data to create baseline profiles following the removal of excessive noise and aberrations; the identification of temporal effects; prospective evaluation of detection algorithms; and finally real-time monitoring and implementation. Building upon the institutions' existing data management software, all steps to use those data for the purposes of syndromic surveillance were set up using open source software; as a result this approach could be readily adopted by other institutions. Relatively straight-forward development and maintenance is expected to lead to the incorporation of these systems into each institution's surveillance processes, becoming an indispensable tool for diagnosticians and epidemiologists, as well as stimulating further technical development of such systems.

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1. Introduction

At the turn of the millennium Doherr and Audige (2001) pointed out the changing demands in animal health surveillance, as disease control and eradication around the globe have increased the demand to deal with rare events and provide evidence of disease freedom. The authors highlighted their role in the early detection of emerging (or re-emerging) diseases, calling attention to the need to develop and implement “scientifically based approaches

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that use the resources (and data) available". Over the intervening decade technological infrastructures and the health data available within them have developed rapidly, disease prevalence has reduced for some diseases in many areas, and general awareness of the need for early disease detection has grown as a result of publicity around major disease outbreaks such as pandemic influenza as well as the concern regarding bioterrorism events (Buckeridge et al., 2005a; Shmueli and Burkom, 2010).

This has led to rapid developments in the field of syndromic surveillance. In human health, syndromic surveillance has been used not only for the early detection of diseases, but also for real-time monitoring of outbreaks (situational awareness (Fricker, 2011a), monitoring of disease trends, and to provide reassurance of disease freedom (Henning, 2004; Katz et al., 2011). It is a tool to extract surveillance information from continually changing health data sources, in as timely a manner as the rate at which data are gathered in electronic formats. Timely collection and computerization were among the reasons highlighted by Ma et al. (2005) to consider the use of laboratory order data in syndromic surveillance, together with the fact that such data often exhibit high population coverage. The scarcity of centralized electronic collections of clinical data in veterinary medicine further motivates the use of laboratory data in animal health surveillance (Dórea et al., 2011).

After a decade of syndromic surveillance development, the focus today is on developing holistic biosurveillance systems which work in parallel – rather than as substitutes – to help inform and complement other types of surveillance (Fricker, 2011a; Buckeridge, 2011). Authors, who in the last decade researched the potential of data sources and the use of different aberration detection algorithms, now highlight the need to continue research into system design and implementation. They also point out the need to compare methods developed under different conditions and to document their performance, fomenting the sharing of experience in order to enhance syndromic surveillance utility (Fricker, 2011b; Morse, 2012).

The objective of this paper is to present a structured approach to the design and implementation of syndromic surveillance systems using laboratory test request data, based on research at two animal health laboratories with high population coverage. These laboratories were the Animal Health Laboratory in the province of Ontario, Canada and the laboratory at the National Veterinary Institute in Sweden. The latter system is still in its development phase, but comparisons between the two data sources will be made as well as an assessment of the portability of the system and the challenges of working with each data source. The focus of this paper is not on the results of each system particularly, but on the development process.

2. Methods

An overview of the process (assuming the availability of animal health on a structured tabular format, such as laboratory data) is presented schematically in Fig. 1. Details of each step are described below with references for more in depth information. The two syndromic surveillance systems described here were developed in order to achieve

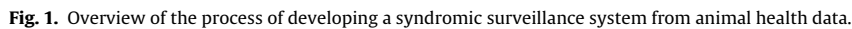
the following goals. Firstly, to generate automated analyses of health data and to generate timely reports containing epidemiologically relevant information; and secondly, to provide a system for early detection of temporal changes in disease patterns that can be indicative of disease outbreaks.

2.1. Data sources

Two syndromic surveillance systems were developed on the initiative of the data owners, with all phases of data access, analysis and output generation developed inside the institution. The Animal Health Laboratory (AHL), part of the University of Guelph, is a full-service veterinary diagnostic laboratory that serves livestock, poultry and companion animal veterinarians in the province of Ontario, Canada. The AHL is the primary laboratory of choice for diagnostics in the province of Ontario. The Swedish National Veterinary Institute (SVA) is an agency under the Ministry of Rural Affairs and is the largest and only official laboratory in veterinary medicine in the country. Both laboratories provide diagnostic services to veterinarians, but also process samples collected as part of monitoring and surveillance programs, as well as commissioned research. Test requests are digitalized daily. At both locations, as in most diagnostic laboratories, data and operations are managed by a laboratory information management system (LIMS), the primary functions of which include reporting the results of diagnostic tests as well as administrative tasks, such as billing. The format of this information is therefore very structured. Most data fields which can be used for surveillance purposes – such as samples submitted and diagnostic tests performed – have a pre-set list of options and only limited amounts of free-text are entered.

These data can be extracted into two-dimensional tables using Structured Query Language (SQL) queries. In both projects a batch of data queried using the institutions' existing data management systems interface sufficed for all development phases. Four years of historical data were used in both cases (2008–2011 for the AHL, and 2009–2012 in Sweden). Only cattle data were used at the AHL, while data from all species are being included at the SVA.

The institutes' diagnosticians were involved in defining the data fields relevant for the extraction of surveillance information. Only data available at the time of submission were considered. Fields with information concerning animal owners and veterinarians were not included in the data extracted for reasons of confidentiality, however any fields that allowed for identification of different animals or herds within the same submission were considered. All fields that could contribute to the identification of the clinical problem were identified. Only cases submitted to the laboratories by field veterinarians requesting diagnostic tests were considered relevant for the purpose of syndromic surveillance, that is, samples submitted as part of regular and/or mandatory surveillance programs were not considered relevant as indicators of the disease burden in the population (Gibbens et al., 2008).



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effects, all of which were explored, namely: time plots; moving average charts and moving standard deviation charts; autocorrelograms; and summary statistics for the overall data and grouped by day of week, month and year. A detailed description of the steps involved in these retrospective analyses is given in [Dórea et al. \(2012\)](#).

2.3.2. Cleaning noise and historical outbreaks

To detect temporal aberrations in real-time, observed data can be compared to historical data known to be free of outbreaks (data-driven methods) or to models reproducing the expected behavior (model-driven methods) ([Yahav and Shmueli, 2007](#)). In both cases, a baseline representing the expected behavior in the absence of outbreaks is necessary. [Tsui et al. \(2001\)](#) proposed a method for cleaning historical data based on the removal of observations above the upper limit of a 95% confidence interval. A key development goal in both systems was to generalize the method so that they could be applied to time series with different statistical properties.

2.4. Prospective analysis

2.4.1. Algorithms for detection of temporal aberrations

For comprehensive reviews of the use of temporal detection algorithms in biosurveillance refer to [Buckeridge et al. \(2005a\)](#), [Shmueli and Burkom \(2010\)](#), [Yahav and Shmueli \(2007\)](#), [Burkom and Murphy \(2007\)](#), [Höhle et al. \(2009\)](#) and references therein. Classical quality control methods have been used extensively in biosurveillance ([Benneyan, 1998](#); [Woodall, 2006](#)). Control charts are based on the assumption that observations are independently drawn from pre-specified parametric distributions, and therefore pre-processing of health data is necessary in order to remove the temporal effects identified ([Lotze et al., 2008](#)). Three control chart approaches were explored: Cumulative sums (CUSUM), Exponentially Weighted Moving Averages (EWMA), and Shewhart control charts ([Woodall, 2006](#)). Holt-Winters exponential smoothing, a data-driven method which can efficiently account for temporal effects, was also investigated ([Burkom et al., 2007](#); [Elbert and Burkom, 2009](#)).

The performance of each of these detection algorithms can be impacted by the specific baseline behavior of each time series ([Buckeridge, 2007](#); [Shmueli and Burkom, 2010](#)), therefore each syndromic group has to be evaluated individually. A minimum of two years of historical data were used to train the algorithms used in these systems, while the remaining data were used to simulate forward monitoring, in order to qualitatively evaluate the performance of different algorithms in the detection of temporal aberrations as well as in the generation of false-alarms.

2.4.2. Define detection limits and evaluate algorithm performance

The main parameter affecting detection of temporal aberrations is the detection limit set for the algorithm. For control charts this detection limit is based on standard deviations, while for the Holt-Winters exponential smoothing algorithm it is based on the confidence interval

of a model prediction for the current time point. Detection limits should be determined in a process similar to the definition of cut-off points for diagnostic methods: smaller detection limits will result in more sensitive diagnostic outcomes, but also in a higher number of false-positives; while limits which are too liberal will miss aberrations. In order to construct standard receiver operating characteristic (ROC) curves, and define detection thresholds based on the desired balance between sensitivity and specificity, the various algorithms were tested using simulated data with controlled injections of outbreaks of various types and durations. The use of simulated data also allowed for comparisons in the performance of several algorithms.

2.5. Real-time monitoring

When implemented in real-time, detection algorithms work in a “roll-forward” manner ([Fienberg and Shmueli, 2005](#)): a fixed-size window of time points (the baseline) is used to “train” the algorithm as to what constitutes “normal” behavior for the data, that is, to calculate the expected value for the current time-point. The current time-point is then read, compared to the expectation calculated and a decision is made concerning the presence of an aberration. It is recommended that a “guard-band” gap is left between the baseline being used to estimate expected behavior and the current time point, in order to avoid the situation in which slowly rising numbers of cases, not sufficient to be detected as an outbreak, will contaminate the baseline ([Burkom, 2003](#); [Buckeridge et al., 2005b](#)). The use of roll-forward algorithms was simulated to a start date at least one-year prior to the time of implementation, in order to evaluate system outputs and to tune algorithm parameters.

2.6. Implementation

Both systems had the goal of developing an application capable of performing daily, automated analysis without the need of human input, without becoming excessively resource intensive. The target platform was a desktop computer on which the system would run and from which a designated analyst could gain access to the analysis and results.

Only freely available software was used. The software RapidMiner 5.0 (Copyright 2001–2010 by Rapid-I and contributors) was used to establish a connection to each institution's database remotely, apply the SQL query to extract data, and execute the text mining and classification process. R code (executing within the RapidMiner interface) was used to perform all statistical analysis, detect alarms, and generate outputs.

3. Results

In this section specific outcomes from the development of the two systems are reported for each of the implementation steps, and where appropriate comparisons between these systems are also noted.

3.1. Data source

Data from AHL contained a specific data field to inform when sampling was part of a monitoring program (active sampling), submitted by a veterinarian for diagnostic (passive submissions), or part of research. In contrast, the main challenge in extracting data at SVA was that of differentiating passive from active sampling and research. Sweden is free from many animal diseases generally regulated, and a large number of surveillance programs are in place (National Veterinary Institute (SVA), 2011). In 2011 40% of all submissions were the result of non-passive activities (monitoring or research), and these submissions represented 71% of all samples processed at the institute in that year. Since the LIMS was not designed for surveillance purposes, a clear definition of the reason for sampling was not always available. An attempt was made to differentiate passive from active sampling based on the party billed. Samples paid from government sources were considered to be part of mandatory programs. Some governmental funding can however be available as an incentive for submission of samples from sick animals. As much as possible keywords that would ensure the observation of a clinical problem in the field – such as keywords related to abortion (abortion, aborted, fetus, placenta, etc.) or words such as “suspicion” or “notification” – were used to find and label samples that should be considered a result of passive submission, regardless of the payer. Conversely, some monitoring programs are paid by the owners, such as certification programs. Experts involved in the programs helped identifying specific laboratorial test codes associated with these activities.

Missing information on animal species also hindered data extraction. Once again keywords search was used in order to look for information among all other fields available at submission. For instance, words like “bovine” and “swine” were searched so that the species could be determined when the diagnostic test was such as “bovine leukemia” or “classical swine fever”. Six percent of all samples within the four years of data extracted had missing species identification. Using the keywords search it was possible to reduce the number of non-identified samples to 1%.

3.2. Automated identification of cases and syndromes

Individual tests are reported as unique data entries. In the AHL database a single submission code is assigned to samples submitted from the same herd in a given day, but the number of animals sampled is not recorded. For this database, therefore, a case was defined as one unique submission. At SVA it was possible to keep track of the number of animals and herds sampled per submission. The definition of case was therefore adjusted for individual syndromes. For abortion in cattle, for instance, a case was defined as each individual animal from which samples were submitted, while for respiratory syndromes the herd was the epidemiological unit of choice. These choices were based on the opinions of the diagnosticians and epidemiologists involved in these projects, but the systems were

designed to record the number of herds and of animals, and therefore the definition of cases can be adjusted as needed.

At AHL, all the historical data available (for cattle) were reviewed manually in order to identify the list of syndromes that would exhaustively classify all records. Due to the limited amount of clinical information, the final syndromic classification was based on organ systems targeted by the veterinarian for diagnosis. In order to automate the classification process, a direct correspondence (mapping) was established between pathogen specific tests and syndromes (for instance rabies to nervous syndrome). In case of non-specific tests, such as “bacteriological investigation”, or “histology”, text mining methods were used to search the fields describing the specimens submitted. The process is described in details in (Dórea et al., 2013a). The highest classification performance was achieved when a set of rules linking keywords and syndromic groups was defined manually (rule-based classification) – 99.4% recall and 100% precision. Remaining errors were due to data quality issues, such as misspellings or use of ambiguous acronyms. This was found to be one of the most time-consuming steps of developing the system. The use of text mining and classification based on keywords offers the advantage that new test names introduced in the laboratory after the implementation of the syndromic surveillance system can be correctly classified, assuming that they are likely to contain words that the system is already able to recognize. However, the use of a rule-based method with set rules has the disadvantage that the classifier cannot learn new rules in an unsupervised manner. In this circumstance, the system should be evaluated periodically (say, every 2–3 years) to ensure that the accuracy of the classifier has not declined.

When proceeding to develop a similar system at SVA, little could be borrowed from the previous system. Apart from the obvious language barrier, essential differences existed in the way tests and specimens were recorded. At SVA test codes are more often modified, and test names can be used more freely. For that reason, the first step of classification could not be fully performed by mapping, and rule-based classification was needed also for this step, requiring an extra dictionary of keywords and set of rules. “Research” or “Internal fixes” were added as classification groups in order to clearly find and filter out tests related to these issues. When test names still could not be associated with any syndromic group, automated classification based on specimen submitted depended on having at least a general recognition of test types. For instance, the specimen “blood” does not carry any information regarding the affected organ system if submitted for serological testing, but it could indicate a generalized clinical condition if submitted for bacteriological investigation. Classification routines were therefore created to differentiate virology from bacteriological tests, searching for keywords or the department assigned to process the sample.

In both systems syndromic classification was performed for each individual database entry (test request). Since classification is performed with such a limited amount of information, no attempt was made to try to define a “main syndrome” per submission, and each case was counted in as many syndromes as the organ systems identified. In

Table 1

Syndromic groups defined for monitoring in both systems developed.

	AHL			SVA ^a			
	Cattle	Ruminants ^b	Swine	Equine	Pets ^c	Wildlife	Avian
Clinical syndromes							
Circulatory, hepatic and hematopoietic	X	X	X	X	X	X	X
Gastrointestinal (GIT)	X	X	X	X	X	X	X
Integumentary		X	X	X	X	X	X
Mastitis	X	X	X	X	X		
Musculoskeletal		X	X	X	X	X	X
Nervous	X	X	X	X	X	X	X
Reproductive	Monitored as one syndrome	X	X	X	X	X	X
Abortion		X	X	X	X	X	
Respiratory	X	X	X	X	X	X	X
Systemic	X	X	X	X	X	X	X
Urinary	X	X	X	X	X	X	X
Others	X	X	X	X	X	X	X
Other syndromes created to provide exhaustive classification of records:							
Antimicrobial resistance		X	X	X	X	X	X
Biochemical profile	X						
Other clinical pathology tests ^d	X						
Nonspecific tests ^e	X	X	X	X	X	X	X
Whole carcass ^f		X	X	X	X	X	X
Toxicology tests	X	X	X				
Sub-groups created to avoid over-representation of specific tests within a syndromic group:							
Bovine Leukemia	X						
Bovine Viral Diarrhea	X	X					
Johnes disease	X						
<i>Neospora caninum</i>	X						
GIT-Macroparasites		X		X	X		
GIT-Microparasites		X			X		
GIT-Bacterial diseases		X	X		X		
GIT-Viral diseases		X			X		
Cyathostomins				X			
<i>Taylorella sp</i>				X			
<i>Sarcoptes sp</i>					X		

^a Fish are also monitored. In fish the syndromes monitored are: Integumentary, Respiratory, Systemic, Urinary, and a sum of all cases not fitting in any of these categories.

^b The syndromes are the same for all ruminants, but cattle, small ruminants and wild ruminants are monitored separately.

^c The syndromes are the same for all pets, but cats and dogs are monitored separately.

^d Tests such as total protein, glucose, Calcium or AST.

^e Not classified into any other category.

^f Whole carcass is a generic classification for all animals found dead and submitted whole for necropsy. These are counted independently, that is, classification in any other group does not exclude a case from being identified as a whole carcass.

both systems, groups with small number of cases were merged when possible. Conversely, some specific tests were assigned to an individual monitoring group in order to avoid their high counts to mask small variability in the demands for other tests in the same group. For instance, in the AHL system the group “Bovine Viral Diarrhea” was created because including diagnostic test requests for this disease in the grouping of gastro-intestinal syndromes would account for more than half of the records. In the system developed at SVA gastro-intestinal syndromes were separated into bacterial tests, virology tests, macroparasites and microparasites.

The final syndromic groups for both systems are listed in Table 1. The absence of some syndromic groups in either of the systems was a result of there not being enough test requests of that type in the existing data. Abortion was monitored within the reproductive syndrome at AHL, but the research of methods to monitor rare event syndromes at SVA resulted in abortion cases being monitored independently.

3.3. Retrospective analysis

Once all data were classified, time-series with the number of cases per day, for each syndromic group, were constructed (see Fig. 2 for examples from both systems).

3.3.1. Evaluation of temporal effects and series modeling

All time-series in both systems showed strong day-of-week effect. The syndromic series at AHL in general had low daily counts (medians from 0 to 9), and showed weak seasonal effects and no global trends as demonstrated in an extensive retrospective analysis documented in Dórea et al. (2012). All series with median counts greater than one case per day were successfully modeled using Poisson regression with month and day-of-week as predictors (Dórea et al., 2012, 2013b). Syndromes with counts lower than that were aggregated into weekly counts, and the resulting time series were not found to exhibit temporal patterns that could impact the performance of outbreak detection methods. The SVA data produced a much larger number of

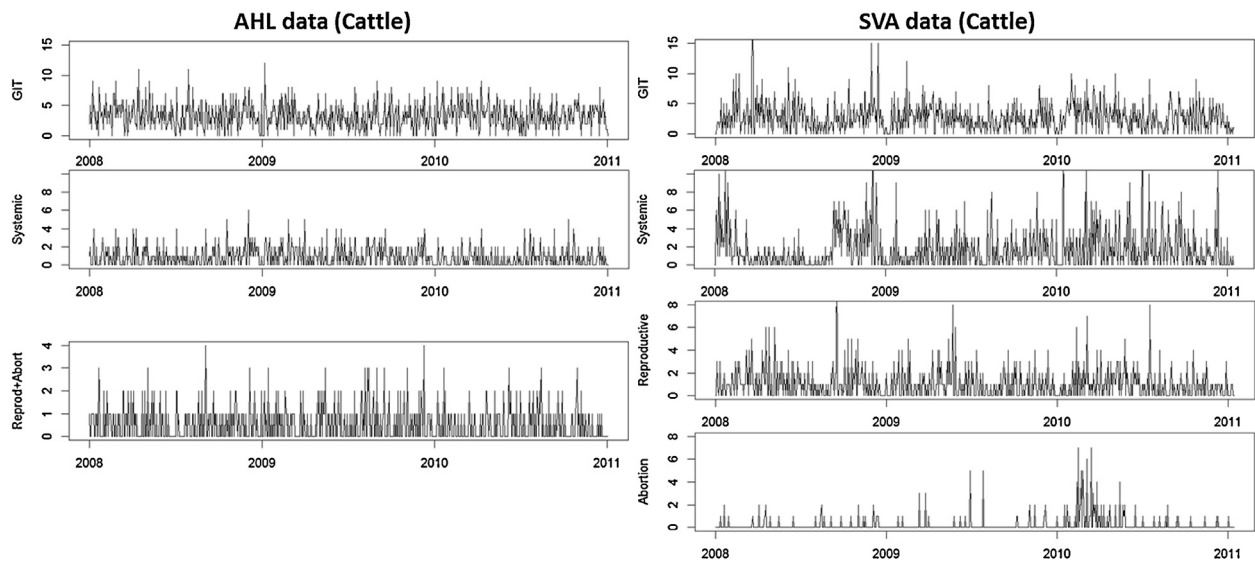


Fig. 2. Example of syndromic time-series from the two systems developed. Plots show the daily number of tests requests to each syndrome.

syndromes, some with daily medians as high as 45. These time series showed a greater variety of temporal effects. Clear seasonal patterns were observed even in series with very low counts, especially for equine submissions.

A greater number of syndromes with very low daily counts were found in the SVA data, and in some cases even weekly aggregation would result in a large number of time points with zero observations. Abortion in cattle, shown in Fig. 2, is one example. Methods were sought to deal with these sparse counts, and to generalize the methods for use with any syndromic series. Three options are being evaluated, as exemplified in Fig. 3 (Lucas, 1985): (i) using distributions appropriate for over-dispersed counts, such as negative binomial (Zeileis et al., 2008; Sparks et al., 2010b); (ii) monitoring the time between two submissions (in days) (Radaelli, 1998); and (iii) monitoring the total number of submissions between each submission for a specific syndrome (Sego et al., 2008; Szarka and Woodall, 2011). The latter has the advantage of correcting for an overall increase in the number of submissions to the laboratory. Statistics to monitor low count data can also be based on monitoring the proportion of overall tests that are classified into the class of interest (number of test requests for the syndrome of interest divided by the total number of tests requests) (Wu et al., 2011). This was not considered an option because these approaches rely on counting the proportion at fixed time intervals (for instance monthly), which was not considered to fit the purpose of early detection.

Options (ii) and (iii) greatly reduce the number of time points to be monitored, therefore monitoring of daily counts will be prioritized, and only rejected if goodness-of-fit tests and analysis of residuals give reason to suspect of lack-of-fit for negative binomial distributions.

3.3.2. Removing noise and historical outbreaks

Using the AHL data it was possible to generalize the method proposed by Tsui et al. (2001) using Poisson

regression, as previously documented in (Dórea et al., 2012). In summary, a Poisson model was fit to the syndromic time series, values above the 95th percentile interval were substituted by the limit of that percentile, and the process was repeated iteratively until no observations fell outside the limit. For series grouped into weekly counts the process was based on the 95th percentile of observations in a 26 weeks moving window.

With the much larger number of time series generated in the SVA system, it was possible to generalize the process of data cleaning to two alternative methods: (i) when temporal effects are found in the data, find a regression model that fits the data, and then apply the iterative steps of model fitting and removal of outliers described in the previous paragraph; or, when no reproducible temporal effects need to be removed, (ii) apply the non-parametric procedure based on moving percentiles.

So far it was possible to prove that these methods can be generalized when the time-series can be successfully modeled using Poisson or negative binomial regression. Zero-inflated versions of these models are available in R (Zeileis et al., 2008), but for the data explored in this study they were not found to represent a gain in fit (Dórea et al., 2012). For time series with strong seasonal effects the use of harmonic and auto-regressive terms as proposed by Burr et al. (2006) and Sparks et al. (2010b,c) are being investigated.

3.4. Prospective analysis

3.4.1. Algorithms for detection of temporal aberrations

During development of the AHL system, it was found that Holt-Winters exponential smoothing, previously suggested as a pre-processing method to remove temporal effects (Yahav and Shmueli, 2007), performed better when used directly as an aberration detection algorithm. Each time series was also subjected to weekly differencing, which removes the day-of-week effect, and the resulting

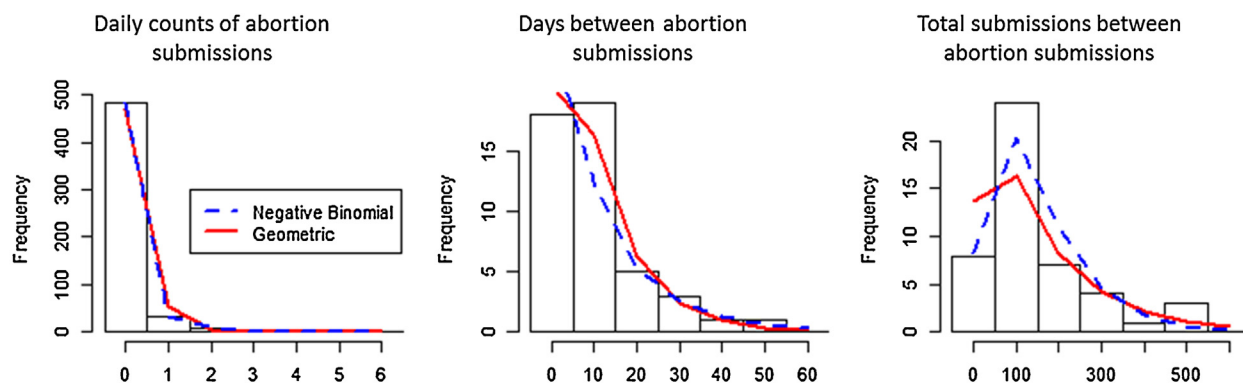


Fig. 3. Three different ways of finding an appropriate distribution for sparse count data, exemplified by the submissions classified as abortion cases in cattle, in the SVA data, from 2008 to 2009.

series was then subjected to aberration detection using control charts. Exponentially Weighted Moving Averages (EWMA) and Shewhart control charts showed complementary performance in detecting outbreaks of different shapes, while cumulative sums (CUSUM) charts did not seem to improve performance once the first two control charts were already in use. Parameters for the detection algorithms were evaluated in order to achieve robust performance for all the syndromes tested. The process of data pre-processing and algorithm choice has been documented in details (Dórea et al., 2013b, 2013c).

When developing the system at SVA, only minor adjustments were required to generalize the system built at AHL. The use of the Holt-Winters algorithm, complemented by the use of control charts applied to data pre-processed in order to remove temporal effects was also shown to represent the best detection approach. The removal of temporal effect was however, more challenging, as stronger seasonal effects were found. Therefore, instead of differencing, pre-processing was based on the use of negative binomial regression models. The relevant regression variables were defined during the retrospective analysis. During prospective analysis, the model is refit to the last two years of data available. Residuals of this model are then used as the time-series to present to control charts for detection.

One essential methodological difference is that SVA surveillance officials are interested in monitoring not only increases in syndromic counts, but also decreases. The package {qcc} in R provides control charts with two-sided detection limits (Scrucca, 2004), but methods for discrete data have not been implemented in the package yet. The use of control-charts with count data will have to be evaluated in the future.

3.4.2. Define detection limits and evaluate algorithm performance

At AHL, evaluation using simulated data showed that no single algorithm performed better in all outbreak scenarios (Dórea et al., 2013b). EWMA charts, for instance, showed low sensitivity for single spikes in the number of counts, or flat outbreaks, but the highest performance for slow raising outbreak shapes (linear, exponential and lognormal) for the mastitis series, which daily median number of

submissions is nine. For series with lower medians, Holt-Winters performed better in exponentially increasing outbreaks. Shewhart charts were often less sensitive than other algorithms, but when they did detect outbreaks, the time to detection was shorter (faster detection).

In order to construct a system that takes advantage of the strengths of each detection algorithm, EWMA and Shewhart control charts, and the Holt-Winters exponential smoothing were combined into a scoring system. For each of these three algorithms five detection limits were applied, using scores from 1 (most sensitive) to 5 (most specific). In the event of an observed count being higher than any of these five detection limits, a score is registered corresponding to the highest detection limit exceeded. As a result, the system outcome for each syndrome, at each time point monitored, is a detection score between 0 and 15 (the maximum sum of the scores from all three algorithms), providing a progressive scale for the magnitude of any alarm. Analysts can tune the system by defining, individually for each of the syndromes, the score threshold that will trigger a system action (alarm). This process is documented in detail in (Dórea et al., 2013c). When applied to simulated data, the scoring system showed high sensitivity for detection of outbreaks, but also generated a high number of false alarms. However, when applied to real data (AHL data), no more than one false alarm per month was documented. Evaluation using simulated data did not replace the need to tune the system using real data, and the score threshold for each syndrome had to be determined individually, using real data.

Based on these conclusions, the process is being reviewed in order to make the definition of alarms more scalable to the size of the system being developed at SVA. Sparks et al. (2010a,b,c) presented methods based on Poisson regression and negative binomial regression fitted to data in moving windows (transitional models), which can be applied to time-series from very low median daily counts to series with higher counts (the authors reported evaluation of series with expected counts from 0.25 to 30 cases per day), and allow more dynamic training and adjustment when several time series must be monitored. These methods are currently being explored.

3.5. Real-time monitoring

During the development of the system in Canada, prospective analysis was performed using data queried as a batch, but applying roll-forward algorithms. Only after the system had evaluated the full amount of historical data available actual querying of data in real-time was set up.

The use of a guard-band was found to prevent baseline contamination only before the first detection of an outbreak, but not after. Once algorithms were implemented in a roll-forward routine, Holt-Winters exponential smoothing was set to correct any aberrations detected, continuously storing an outbreak-free baseline. The goal was to remove aberrations and excess noise, so the correction is applied anytime observed data is greater than the upper limit of a 97.5% confidence interval for the algorithm prediction, independently of the generation of an alarm in the combined scoring system. The observed count is then substituted by the limit of that confidence interval (Dórea et al., 2013c).

3.6. Implementation

The output in case of an alarm is an email to analysts involved in the project, containing summaries of the syndromic series as well as the detection score generated by each of the three algorithms. Every Friday the analysts receive emails with 26 and 52 week plots for all syndromes monitored, regardless of the generation of an alarm during the week. Examples can be found in Dórea et al. (2013c).

4. Discussion

Two systems for the automated extraction of surveillance data from animal laboratory databases have been developed, and comparisons have been described in this paper. In this section we present the lessons learned and provide recommendations for future system implementation.

The collection of specimens for diagnostic purposes happens late in the disease continuum, and covers a small percentage of the population affected. For this reason, sensitivity and timely detection should be prioritized (Widdowson et al., 2003). In both systems developed, the data were monitored directly at the institution where they are collected and stored. A single, centralized, electronic source circumvents the need to gather data from scattered locations, as well as the need to develop standards that allow integration of these data from different sources. Issues of data transfer, sharing agreements and security are all also avoided if the data are monitored directly at the source institution.

The speed of extraction of surveillance information from health data does not depend solely on the rate of data collection, but also on the ability to translate data from the format in which they are collected to a meaningful surveillance output (Fienberg and Shmueli, 2005). A challenge of working with laboratory data was that of determining how to transform diagnostic requests into epidemiological information, especially due to the limited clinical information received with each submission. On the other hand,

due to the structured nature of laboratory data, when compared to clinical narratives, automated recognition of syndromes could be performed using simple methods based on keyword search. It was not necessary, for instance, to deal with contextual information, such as negation, expressions of uncertainty, etc. (Chapman, 2006). The most time-consuming step associated with classification, in both systems, was the creation of a dictionary of keywords relevant to each classification task, and the definition of the relationship between these words, their co-occurrences and the target syndromic group. Once defined, however, these relationships were easily formulated into a set of rules that achieved high classification performance.

Failures in automated classification of records could be tracked to two main problems: the secondary use of data stored for purposes other than surveillance – in the specific cases described here, for diagnostic purposes and management tasks; and data quality. Involving diagnosticians, as well as the technicians responsible for digitalizing the information daily, helped inform how best to mine surveillance information from data organized primarily for billing and management. It also helped elucidate issues related to data quality, and the reasons behind missing or erroneous information. Small artifacts can hinder the detection of cases by the automated classification algorithms, and result in large impacts on the performance for an entire system (van den Wjingaard et al., 2011). As these systems are developed, feedback of these issues to the institution can help increase overall data quality. There is often a ‘disconnect’ between the people involved in data generation – laboratory staff – and those making use of the system outputs – epidemiologists. In order to maintain the involvement and enthusiasm of the first group, it is necessary for them to be involved in the process, understand the benefits of the system, and possibly also gain insights from system implementation (Richards et al., 2010).

One of the recommended parameters to evaluate in a surveillance system is its portability, that is, “how well the system could be duplicated in another setting” (Centers for Disease Control and Prevention (CDC), 2004). In the systems described here the steps to achieve syndromic classification showed very limited portability, but once all data were reduced to multiple time-series little difference existed between the steps necessary to handle data in the two different systems. The methods developed were highly portable, but applicable only to series with the limited range of temporal effects observed in the AHL data. Building on that experience, methods to deal with a broader range of temporal effects are being developed, thus generalizing the system still further. Once temporal effects have been individually identified and modeled or removed, prospective steps implemented at AHL will be readily transferrable, as shown in preliminary analysis. Once again, however, individual evaluation will be required in order to parameterize the algorithms and to adjust reporting thresholds to achieve maximum sensitivity and specificity.

During current system development at SVA more effort is being applied to the monitoring of sparse data, avoiding the need to group syndromic groups or decrease the frequency of monitoring (to weekly for instance). This is due to the opportunity to build on the knowledge

developed in the first system, the recognition that the system has to be robust for a larger number of syndromic groups, and also due to the different surveillance goals of each institution. Sweden is free of many reportable diseases, which increases the number of rare syndromes to monitor. Moreover, outputs from the system may be used to document disease freedom, and the aggregation of syndromic groups into broader categories is likely to reduce the specificity of syndromic monitoring. Disease freedom documentation using laboratory data has been described in Australia (Richards et al., 2010).

The large number of surveillance activities in Sweden also resulted in an interest to monitor decreases, as well as increases, in the rate of laboratory submissions. Decrease in passive submissions could be caused by reduction of the catchment area of SVA (possibly due to professionals sending diagnostic requests to other laboratories, or reduced awareness in the field), which is important to notice and drive efforts to correct, in order to retain the population under surveillance by the system. Decreases in active surveillance are also relevant in terms of the need to detect and correct.

In both systems the indicators monitored are the number of laboratory submissions for specific syndromes; denominator data are not considered. This assumes some consistency in the rate of laboratory submissions in the catchment population over time, an assumption common in syndromic surveillance systems (Shaffer et al., 2007). An increase in the rate of laboratory submissions would be detected by the monitoring systems, which was considered a desired feature by both institutions. A decrease in the total number of submissions could occur if animal owners and veterinarians reduced the rate of laboratory submissions for diagnostics, or had the option of submitting to other laboratories. However, such reductions would not be expected to happen suddenly, and it is likely that the detection algorithms would adjust to the new baselines.

Aiming at adjusting to the larger scale of the new system being developed at SVA, methods to achieve fast set up of detection thresholds are being researched, such as mathematical estimation of specificity, as opposed to the large scale simulation studies. The results may, in practice, not precisely reflect the theoretical values, but the use of a scoring system gives the system enough flexibility for parameters to be tuned to real data after implementation. Once the system is fully developed, it can be implemented retroactively to two years prior the actual implementation date, allowing review of what would have been the system's output for that period, in order to customize the system syndrome by syndrome. This step was found to be essential during development at AHL. By automating the definition of the detection limits without the need of large scale simulations, however, the system will be more rapidly ready for this real-time, real-data customization.

The development of a system robust for several syndromes and with a highly flexible definition of what constitutes an alarm represents a trade-off: fine customization is possible, but at the expense of analysts' time to review system outputs and tune parameters. In both institutions training of analysts to read system outputs and take decisions regarding adjustment of detection thresholds is

on-going. As the analysts become familiar with the system, the cycle of feedback between the analyst and the system will both increase performance of the system and feed the institution with surveillance information which goes beyond early disease detection (Widdowson et al., 2003). If time was not invested by the institutions in this task, the full utility of the system would not be achieved.

Animal health data are subjected to several sources of variation not related to diseases, such as economic factors (Alton et al., 2010; O'Sullivan et al., 2012). The low counts for most syndromic groups monitored makes it difficult to model all these factors and take them into consideration when removing temporal effects. It may be necessary to deal with the false-alarms, which is easier when an analyst is constantly reviewing outputs and understands the data for each syndrome. Monitoring directly at the source institution has further advantages for reviewing each alarm. Analysts can seek more information in real time, or follow-up with the samples while they are still being processed at the institution, taking action for immediate response after diagnostic results are available, or even before, depending on how strong the evidence is at the time of submission. The gain in early detection can only be realized if it triggers immediate reaction (Fienberg and Shmueli, 2005).

Both systems were developed as desktop applications at the institution which owns the data, and are scheduled to run on a daily basis. Security is assured because identification fields are never queried, and the data queried is rapidly converted into matrices of daily counts, discarding original data. The structured development and implementation resulted in the swift completion of these projects. Some previous attempts to develop syndromic surveillance in veterinary medicine have failed to sustain engagement and become established systems (Dórea et al., 2011). The relatively straightforward nature of the systems summarized here should facilitate their incorporation into the routine activities of each institution, until they become a basic tool for the associated diagnosticians and epidemiologists (Hurt-Mullen and Coberly, 2005).

5. Conclusion

The development of modern biosurveillance systems is increasing in veterinary medicine based around several decades of experience that have been accumulated in public (mainly human) health. The documented experience indicates that the true utility of syndromic surveillance can only be realized if attention is given to the potential of these systems for more than early disease detection (Fricker, 2011b). Two systems have been developed for the automated extraction of surveillance information from animal laboratory data and have been compared in this paper. We have discussed how the experience accumulated in the development of one system largely facilitated the development of the other system using a completely different data source. It is hoped that the experiences shared in this paper will further encourage and facilitate the development of other system using animal health data. All steps were set up using open source software, as desktop applications, which could be readily reproduced in other institutions. Fast development and simple maintenance is

expected to lead to incorporation of these systems into the institutions' routine, becoming an indispensable tool for diagnosticians and epidemiologists, and fomenting further technical development.

Conflict of interest

The authors declare no conflict of interest.

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